Chapter 10
IGBTs

10.1 Mode of Function

A lot of work was spent to combine bipolar devices with their superior current density with the possibility of voltage-control as given in MOSFETs. Early works tried to combine thyristor-related structures with MOS gate control. However, a transistor-based device won the race. Plummer and Scharf from Stanford University California [Plu80, Scf78] worked on planar MOS controlled TRIACs, they discussed an intermediate bipolar state as a step before the intended thyristor mode. They even mentioned a kink in the characteristics toward better conductivity and showed a measured IGBT-like I–V characteristics. However, the final aim was thyristor mode. The Insulated Gate Bipolar Transistor (IGBT) was invented in the United States of America by F. Wheatley and H. Becke from RCA Corporation in U.S. [Bec80]. The patent clearly points out "Although this creates a four layer device, the conductivities and geometries of the four semiconductor regions are manipulated so as not to form a regenerative thyristor". In 1982, the same year as said patent was published, the first experimental demonstration of a practical discrete vertical IGBT device was reported by Baliga et al. from GE [Bal82], and the advantage of a 10 times higher current density compared and MOSFET and two times compared to the bipolar transistor was shown. Also Russel et al. from RCA submitted a paper describing the IGBT in the end of 1982 [Rus83]. The IGBT was firstly referred to as COMFET (Conductivity Modulated FET) [Rog88, Rus83], also the term IGT (Insulated Gate Transistor) [Bal83] was used. A report on the first story of the IGBT is given in [She15]. A description of the technical details leading to a mature device is made in [Iwa17].

It started an intensive work of several groups on the new device. Nakagawa and Ohashi from Toshiba showed the first 1200 V device [Nak84]. About 10 years later, IGBTs were introduced in the market by manufacturers from Japan and Europe. In a short time, IGBTs won an increasing share of applications, and they

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replaced the formerly used bipolar power transistors, and nowadays even GTO-thyristors in the high power range.

In a first rough approach, the IGBT can be seen as a MOSFET, in which the n^+-layer at the drain side is replaced by a p-layer. Figure 10.1 shows the structure of the IGBT.

The notations collector and emitter have been taken from the bipolar transistor, the notations anode (for collector) and cathode (for emitter) also make sense.

If at the IGBT a positive voltage between collector C and emitter E is applied, the device is in the blocking mode. If now a voltage V_G higher than the threshold voltage V_T is applied between gate and emitter, an n-channel is created; electrons flow to the collector (Fig. 10.1). At the collector side pn-junction, a voltage in forward direction is generated, and holes from the p collector layer are injected into the lowly doped middle layer. The injected holes allow an increased charge carrier density; the increased carrier density lowers the resistance of the middle layer, and the conductivity of the middle layer is modulated. Like in a MOSFET, the turn-on and turn-off of the IGBT happen by the creation and removal of an n-channel by applying a gate voltage. Regarding threshold voltage and channel resistance, the same holds as described in Chap. 9 for the MOSFET.

Figures 10.1 and 10.2 show a pnpn-structure as in a thyristor, however the action of this thyristor is strictly avoided by a high-conductive emitter short R_S realized by a highly doped p^+-layer in the center of the cell [Nak84]. Figure 10.2b shows the equivalent circuit of the four-layer structure. With the pnp and npn partial transistor, a parasitic thyristor structure is visible. The emitter of the npn partial transistor and its base are shortened by the resistor R_S. With this, the current gain of the npn partial transistor is eliminated at low current. But at very high current, the npn transistor can be activated, and the parasitic thyristor can be triggered into the on-state mode

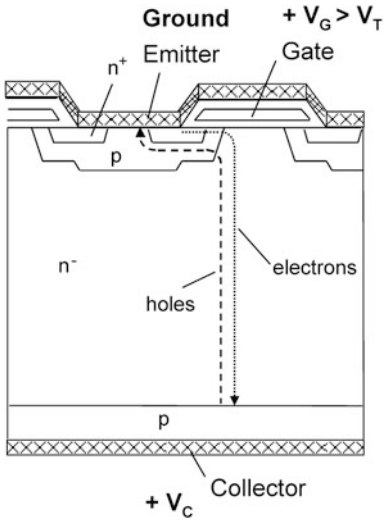


Fig. 10.1 IGBT in conduction mode at $V_G > V_T$, electron and hole current

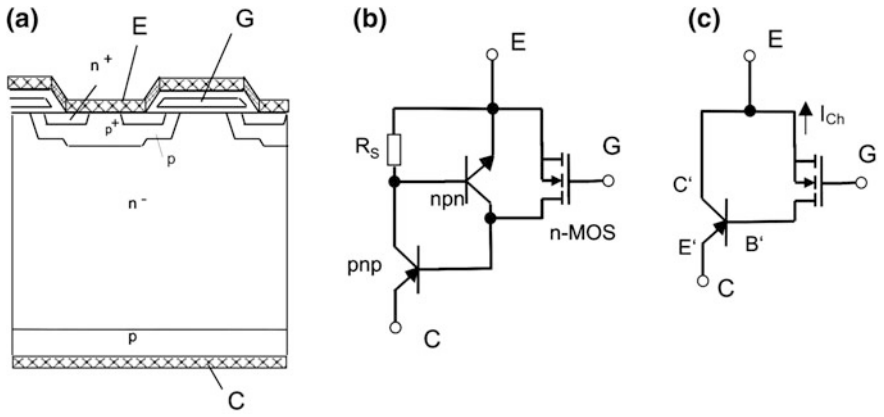


Fig. 10.2 IGBT **a** simplified structure **b** equivalent circuit with parasitic npn-transistor and resistor R_S **c** simplified equivalent circuit

with the internal feedback loop with the partial pnp-transistor. This effect is denoted as latch-up: the device can no longer be controlled by the MOS gate. Latching of the parasitic thyristor is a destructive effect for the IGBT.

For sufficiently low R_S, the npn partial transistor can be neglected, and the simplified equivalent circuit, as shown in Fig. 10.2c, is obtained. This is the most important equivalent circuit for understanding the IGBT. The terminals of the pnp-transistor are denoted with C', E' and B'. The collector C of the IGBT is the emitter E' of the pnp-transistor. Concerning the physics of the IGBT, it is an emitter.

The channel is created for a gate voltage V_G higher than the threshold voltage; in the base terminal of the pnp-transistor the channel current I_{CH} is flowing. For the current $I_{C'}$ at C' then holds $I_{C'} = \beta_{pnp} \cdot I_{CH}$, or, with the relation $\alpha = \beta/(\beta + 1)$ known from the sections on bipolar transistors

$$I_{C'} = \frac{\alpha_{pnp}}{1 - \alpha_{pnp}} \cdot I_{CH} \tag{10.1}$$

For the collector current of the IGBT holds

$$I_C = I_{C'} + I_{CH} = \frac{\alpha_{pnp}}{1 - \alpha_{pnp}} \cdot I_{CH} + I_{CH} = \frac{1}{1 - \alpha_{pnp}} \cdot I_{CH} \tag{10.2}$$

Therefore the collector current of the IGBT is always higher than its channel current. The saturation current of the IGBT will also be much higher than that of the MOSFET. With the parameter of the channel conductivity κ, defined for the MOSFET in Eq. (9.4), results for the IGBT

$$I_{Csat} = \frac{1}{1 - \alpha_{pnp}} \cdot \frac{\kappa}{2} \cdot (V_G - V_T)^2 \tag{10.3}$$

However, α_{pnp} must be adjusted not too high, and to meet all the different requirements, it should be adjusted very exactly. The current necessary for latch-up must be shifted to such a high value which will not occur in application. To achieve this, different measures are used in the design of the IGBT, which will be explained in the following.

10.2 The I–V Characteristic of the IGBT

The I–V characteristic of an IGBT in forward direction is shown in Fig. 10.3. The characteristic has some similarities to that of a MOSFET.

For a gate voltage V_G higher than the threshold voltage V_T, the channel is open. The IGBT differs from the MOSFET by the junction voltage of the additional pn-junction at the collector side. The IGBT is operated in the saturation region, as is usual for power devices and as it was already the case with the MOSFET and the bipolar transistor. The operation point is on the branch of the characteristic for V_G = 15 V. At this branch for operation at a given current I_C, the generated voltage drop V_C is read off.

Figure 10.4 compares the characteristic of an IGBT for V_G = 15 V with that of a bipolar transistor, also at a high drive level of I_B = 2.5 A. Both devices are specified for 600 V and have a comparable area. The threshold voltage caused by the reverse pn-junction is recognizable in the characteristics of the IGBT. At low current densities, the bipolar transistor has the inferior voltage drop because no threshold voltage occurs with it. However, at higher current densities, above 14 A, the forward voltage of the IGBT is much lower than that of the bipolar transistor. The shown IGBT is rated to a nominal current of 20 A, this current level is not reached with the used bipolar transistor even at a high base current.

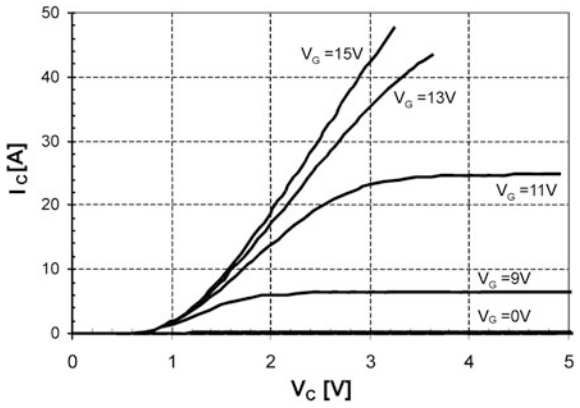


Fig. 10.3 I–V characteristic of a 20 A / 600 V IGBT

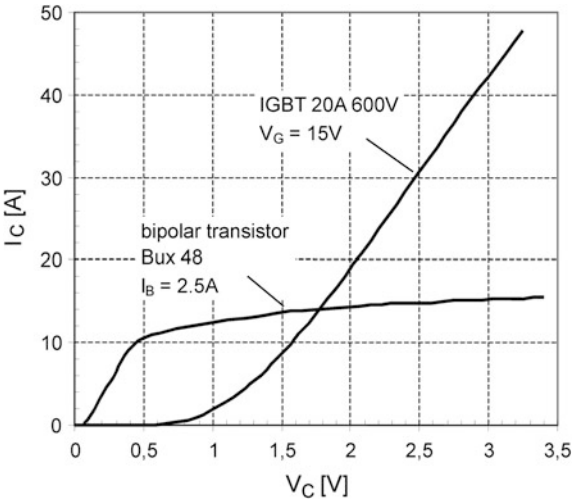


Fig. 10.4 Forward voltage-current characteristic of an IGBT in comparison to a bipolar transistor

A comparison of power devices of a higher voltage range would show the difference even more drastically. The IGBT especially can be designed also for higher voltages; it is not restricted like the MOSFET or bipolar transistor by physical mechanisms. Meanwhile, IGBTs have been fabricated for voltages up to 8 kV [Rah02], they have been commercially available in 2016 for voltages up to 6.5 kV.

10.3 The Switching Behavior of the IGBT

The determination of the switching behavior of the IGBT is done in a circuit according to Fig. 10.5 with inductive load. The time-constant of the load $\tau = L/R$ is chosen so high that courses of voltage and current can be assumed as constant before the switching instant.

The turn-on process of a field-controlled device was discussed already with the MOSFET. The relation between the rise time of the current, the fall time of the voltage, the internal capacities and the chosen gate resistors are the same as were discussed in the context with Fig. 9.18. The IGBT is used with a freewheeling pin-diode in almost all applications; at the turn-on process, it additionally has to take over the reverse current peak and the stored charge of the freewheeling diode. The processes are shown in Figs. 5.20 and 5.21 and are described in context with Eqs. (5.82) to (5.84). For the turn-on energy per pulse in the IGBT, it can be given under the same simplifications

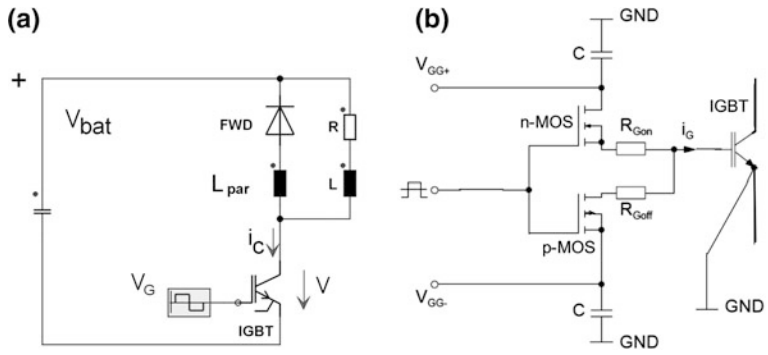


Fig. 10.5 Measurement of the IGBT switching behavior. **a** Power circuit **b** Simplified output stage of a gate drive circuit containing gate resistors R_{Gon} and R_{Goff}. Adapted from [Nic00]

$$E_{on} = \frac{1}{2} \cdot V_{bat} \cdot (I_C + I_{RRM}) \cdot t_{ri} + \frac{1}{2} \cdot V_{bat} \left(I_C + \frac{2}{3} I_{RRM} \right) \cdot t_{fv} \tag{10.4}$$

A more exact determination is done with the oscilloscope, see Eq. (9.33).

At turn-off of the IGBT, the positive gate voltages are put on zero or a negative value, and in the first time interval, processes appear like those described with the MOSFET in the context of Fig. 9.19. As long as the stored charge in the IGBT is not too high, similar relations between the rise time of the voltage, the internal capacities and the chosen gate resistance are valid. If the gate capacity is discharged abruptly, the channel current is interrupted. Usually a gate resistor is applied, and V_G is reduced to the value of the Miller plateau at $V_G = V_T + g_{fs} \cdot I_D$. While the voltage increases, the current in the IGBT in a circuit with inductive load flows on continuously until the voltage is higher than the applied battery voltage V_{bat}. During the voltage rise time, the channel current is reduced and the hole current flowing through the p-well is increased in the same amount. Carriers from the n-base are extracted by the hole current. The hole current is increased compared to the steady-state conduction mode. The problem of possible latch-up is most serious at the turn-off event. The requirement that latch-up must be avoided limits the maximal possible current which an IGBT can turn off.

The hole current leads to the removal of the charge carriers from the n-base, a space charge is built-up, and the device takes over the applied voltage. After the voltage has increased to V_{bat}, the current falls steeply. With IGBTs, the slope of the decreasing current di_c/dt can be adjusted by the gate resistor only with limited effect. The slope di_c/dt causes an inductive voltage peak at the parasitic inductance, and the forward recovery voltage peak V_{FRM} of the freewheeling diode adds to this. The voltage peak amounts to

$$\Delta V = L_{par} \cdot \frac{di_c}{dt} + V_{FRM} \tag{10.5}$$

The higher the nominal voltage range, the more significant the part of the V_{FRM} may be. See Sect. 5.6 especially Fig. 5.15.

As the main difference to the MOSFET and to the bipolar transistor, the IGBT features a tail current at turn-off. A measurement of the turn-off of the IGBT including the tail current is shown in Fig. 10.6. The current falls down to the value I_{tail}, and then it goes down slowly during the time interval t_{tail}. The measurement of the end point of the tail current is difficult since it decreases very slowly. The time of the tail current t_{tail} is determined by the recombination of the remaining charge carriers in the device. With the typical high charge-carrier lifetime in the NPT IGBT, t_{tail} can amount to several µs, while t_{rv} is in the order of some 100 ns and t_{fi} is in the range of 100 ns.

During the time of the tail current, the voltage is high and the losses created in this interval cannot be neglected. In practice, the determination of the turn-off energy per pulse is mostly done with an oscilloscope; the product of current waveform and voltage waveform is executed and integrated over the turn-off time interval. A simplified estimation can be made with

$$E_{off} = \frac{1}{2} \cdot V_{bat} \cdot I_C \cdot t_{rv} + \frac{1}{2} \cdot (V_{bat} + \Delta V) \cdot I_C \cdot t_{fi} + \frac{1}{2} I_{tail} \cdot V_{bat} \cdot t_{tail} \tag{10.6}$$

IGBTs are usually used in bridge configurations. The IGBT is turned off mostly with a negative gate voltage, as in Fig. 10.6. In the blocking mode, a voltage of −15 V is applied, sometimes also a smaller voltage of −8 V. This does not have,

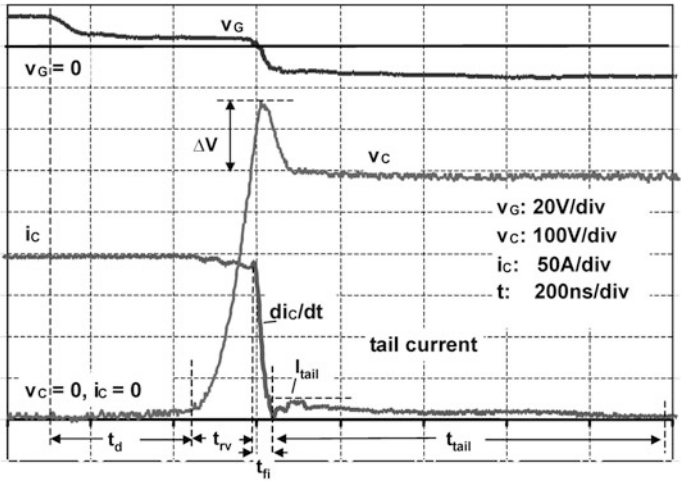


Fig. 10.6 Turn-off of an NPT-IGBT (200 A/1200 V module BSM200GB120DN2, manufacturer Infineon). T = 125 °C, $R_{Goff} = 3.3\,\Omega$

beside of the waveform of the gate voltage, much effect on the waveforms of current and voltage at turn-off.

10.4 The Basic Types PT-IGBT and NPT-IGBT

In the very first IGBT-structures the n^+-substrate of the MOSFET was replaced with a p^+-substrate. These structures were very sensitive to latch-up of the parasitic thyristor. The behavior could be improved by adding a medium-doped n-layer, a so-called n-buffer, between p^+-substrate and lowly doped n^--layer. This n-buffer must have a sufficient doping N_{buf} [Nak85]. The electric field can penetrate into the n-buffer, a trapezoidal shape of the electric field is given. From this characteristic, the notation Punch-Through IGBT or *PT-IGBT* was derived (In the proper meaning of the word, this notation is not correct, see Sect. 5.1). The structure is shown in Fig. 10.7.

As already mentioned, α_{pnp} must be adjusted and must not be too high. The doping of the buffer N_{buf} is of influence on it. In the section on the bipolar transistor, α was composed of two terms

$$\alpha = \gamma \cdot \alpha_T \tag{10.7}$$

For the estimation of the emitter efficiency γ, the Eq. (7.23) can be used; for the given case of a p-emitter it can be written as [Mil89]

$$\gamma = \frac{1}{1 + \frac{\mu_n}{\mu_p} \cdot \frac{N_{buf}}{N_{sub}} \cdot \frac{L_p}{L_n}} \tag{10.8}$$

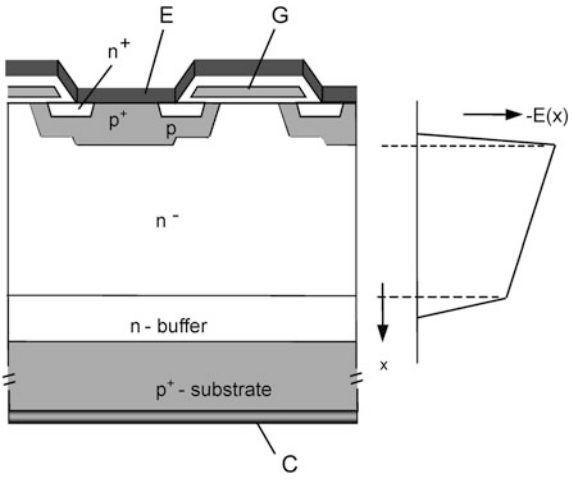


Fig. 10.7 PT-IGBT, structure and field shape

Since L_p, the diffusion length of holes in the n-buffer, and L_n, the diffusion length of electrons in the substrate, are in the same order of magnitude, and since μ_n/μ_p amounts to 2–3, it is difficult to make the denominator in Eq. (10.8) significantly larger than 1, as long as the doping of the buffer N_{buf} is not in the same range of magnitude as the doping of the p^+-substrate N_{sub}. Any control of γ will be difficult. For the PT-IGBT, we can therefore assume $\gamma \approx 1$.

Thus the adjustment of α_{pnp} in the PT-IGBT is done by the transport factor α_T. According to Eq. (7.29), the width of the base-layer w_B and the diffusion length L_p in the substrate are important factors for this[1]:

$$\alpha_T = 1 - \frac{w_B^2}{2 \cdot L_p^2} \tag{10.9}$$

and L_p was given in Eq. (2.118)

$$L_p = \sqrt{D_p \cdot \tau_p} \tag{10.10}$$

With a low carrier lifetime, L_p, α_T and finally α_{pnp} are reduced. For the reduction of the carrier lifetime, the technologies for the creation of recombination centers as described in Chap. 4 are used. Platinum diffusion, electron irradiation, irradiation with He^{++}-ions or protons, or a combination of two of these processes is used; the specific process is different for different suppliers and their special device generations. The devices feature low turn-off losses.

The fabrication of the basic material for PT-IGBTs is done with an epitaxy process. N-buffer and n–layer are deposited on a p^+-substrate. This technology is well applicable in the voltage range up to 600 V. For 1200 V devices, the epitaxial technology demands an increased effort because of the necessary thick epitaxial layer. PT-IGBTs dominated the applications at blocking voltages up to 600 V over several years.

As an alternative concept, the so-called *NPT-IGBT* (Non-Punch-Through IGBT) was introduced. It is based on a suggestion of Jenö Tihanyi [Tih88] and was first realized by Siemens (today Infineon) [Mil89]. The structure is shown in Fig. 10.8. The shape of the electric field is triangular. The device for the same blocking voltage, which is given by the area under the line $E(x)$, must therefore be designed with a much thicker base width w_B. For the very first types of NPT-IGBTs, additionally a relatively high distance between the end of the space charge at $x = w$ and the p-collector layer at $x = w_B$ was chosen. Thus the effective base of the pnp-transistor at high voltage is $w_B - w$. A widened effective base, as expressed with Eq. (10.9), can somewhat reduce α_T and therewith α_{pnp}.

[1]Equation (10.9) is usually derived for low injection. The considerations, however, will also be valid for high injection, qualitatively.

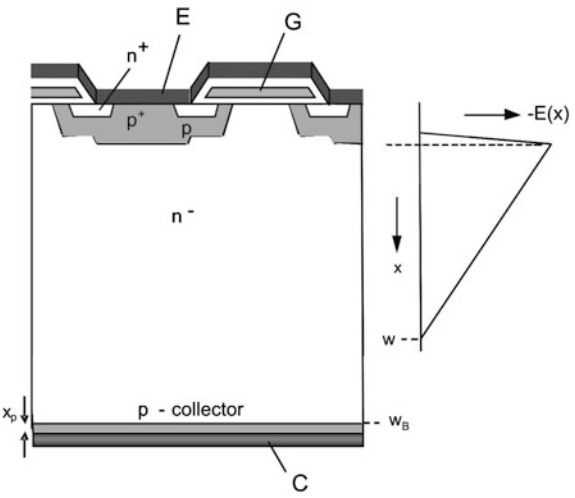


Fig. 10.8 NPT-IGBT. Structure and field shape

The main control of α_{pnp} is done via the emitter efficiency γ. The p-collector layer is lowly doped and its penetration depth is very shallow; with this the low emitter efficiency is adjusted. In Sect. 3.4, the Eq. (3.100) was derived for the emitter efficiency, for p-emitter with $p_L \approx n_L$ it is written as

$$\gamma = 1 - q \cdot h_p \frac{p_L^2}{j} \tag{10.11}$$

To achieve low γ, the emitter parameter h_p must be large. For the p-emitter, it can be expressed by Eq. (3.98)

$$h_p = \frac{D_n}{p^+ \cdot L_n} = \frac{D_n}{p^+ \cdot x_p} \tag{10.12}$$

For small x_p, the diffusion length L_n can be replaced by the very shallow penetration depth x_p of the p-emitter. At low p^+ and $x_p < 1$ µm for the given structure, the emitter efficiency γ will be low. High h_p means a large contribution of the emitter recombination at the total recombination. Special measures to reduce the carrier lifetime are no longer necessary.

The NPT-IGBT is therefore very robust against latch up, and it features high short-circuit robustness [Las92]. A further advantage results from this type of control of the plasma modulation. The temperature dependency of the forward voltage is very suitable for parallel connection. The voltage drop V_C at a constant collector current I_C and a constant gate voltage V_G is increasing with temperature, at the typical operation current range of I_C.

The voltage drop across the lowly doped middle region was estimated for a pin diode by means of Eq. (5.47) as

$$V_{drift} = \frac{w_B^2}{(\mu_n + \mu_p) \cdot \tau_{eff}} \tag{10.13}$$

with the effective carrier lifetime

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{HL}} + \frac{h_p \cdot p_L^2}{w_B \cdot \overline{p}} + \frac{h_n \cdot p_R^2}{w_B \cdot \overline{p}} \tag{10.14}$$

In this Eq. (10.14) the two last terms on the right-hand side stand for the contribution of the emitter regions. Compared to pin diodes, a different profile of the plasma is given for the basic IGBT types (see Fig. 10.8). However, Eq. (10.13) may be used for estimation. For a PT-IGBT structure, the lifetime τ_{HL} dominates the effective lifetime τ_{eff}. τ_{eff} increases with increasing temperature. Depending on the used recombination centers, τ_{HL} increases by a factor of two to four for a temperature increase from 25 to 125 °C. The influence of τ_{eff} dominates in Eq. (10.13), and V_{drift} and therewith V_C decrease with increasing temperature.

For an NPT-IGBT the carrier lifetime τ_{HL} in the lowly doped middle region is adjusted as high, and both terms of emitter recombination dominate in the influence on τ_{eff}. With increasing temperature also τ_{HL} increases, but this is of low effect to τ_{eff}, since the emitter terms dominate, and their temperature dependency is weak. The temperature dependency of voltage across the drift region according to Eq. (10.13) is therefore dominated by the temperature dependency of the mobilities. The mobilities decrease strongly with increasing temperature, it holds $(\mu_n + \mu_p)_{(125\,^\circ\mathrm{C})} \approx 0.5 \cdot (\mu_n + \mu_p)_{(25\,^\circ\mathrm{C})}$. V_{drift} and therewith V_C increase with increasing temperature.

This V_C temperature dependency, also assigned as "positive temperature coefficient of V_C", on the one hand leads to increased conduction losses at high operation temperature. But on the other hand, this is of advantage if devices are connected in parallel: if one of the devices has a lower V_C because of production-induced scattering of V_C and therefore takes more current, its temperature will increase. Then its V_C increases and its current is reduced. With this a negative feedback is given, the system of parallel-connected devices is stabilized.

The manufacturing process of the NPT-IGBT can be controlled more exactly; the adjustment of the emitter efficiency at the collector side can be executed very accurately with ion implantation technology. The NPT-IGBT has become the dominating device in application, because of its robustness and because of the suitable behavior at parallel connection. Meanwhile, also NPT-IGBTs for 600 V have been developed and established in the market.

At turn-off, the NPT-IGBT has a long tail current, see Fig. 10.6. The PT-IGBT has a shorter, but higher tail current. This will be more understandable after the internal plasma distribution has been considered.

10.5 Plasma Distribution in the IGBT

Using the example of the NPT-IGBT, the internal distribution of the charge carriers, the plasma distribution, shall be investigated. Figure 10.9 shows a simulation of this shape of the plasma in the on-state mode for a 1200 V IGBT at different forward-current densities. The lowly doped n-base of the IGBT is flooded with free carriers. Because of neutrality, $n \approx p$ holds for the bipolar device. The shape of the hole distribution is therefore almost identical to the electron distribution shown in Fig. 10.9. The cell structures are on the left-hand side in Fig. 10.9; the vertical coordinate x is the same as in Fig. 10.8. The right-hand side p-collector layer has a penetration depth of less than 1 µm for the NPT-IGBT, it cannot be recognized.

Compared to the plasma distribution of a diode (see Fig. 5.6), the plasma distribution is strongly reduced at the side of the cell structures for the conventional IGBT. It corresponds to the plasma shape in a pnp-transistor; the collector of the IGBT in this case is the emitter of the pnp-transistor, compare Fig. 10.2. In a transistor, typically the plasma density decays from the emitter side to the collector side, see Fig. 7.6 for comparison.

Considering the rated voltage of 1200 V, the figure shows a base width w_B of 250 µm which is very wide. The first NPT IGBTs have been designed with w_B = 220 µm. According to Eq. (5.1) and the respective Fig. 5.5, a base width of 110 µm would be sufficient for the condition of a triangular field shape. However, a very wide n-base was typical for the first generations of NPT-IGBTs.

At turn-off with inductive load, the device must first take over the voltage, while the load current is still flowing. The development of the internal plasma at the turn-off process is shown in Fig. 10.10.

Up to the time t = 0.69 µs the voltage increases up to the value of the applied battery voltage. During the voltage increase, the charge in the left part of the base is extracted quickly. After the battery voltage is reached, the current decays steeply

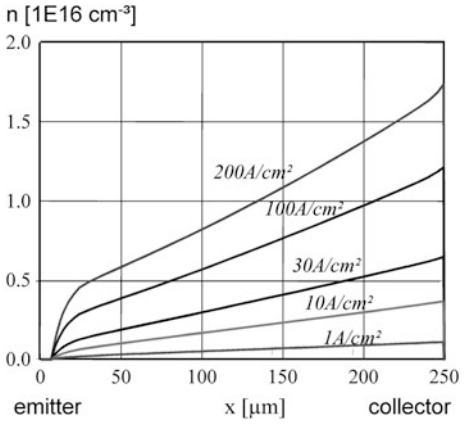


Fig. 10.9 Electron density at on-state, 1200 V NPT-IGBT. Figure from [Net99] © 2008 isle Steuerungstechnik und Leistungselektronik GmbH

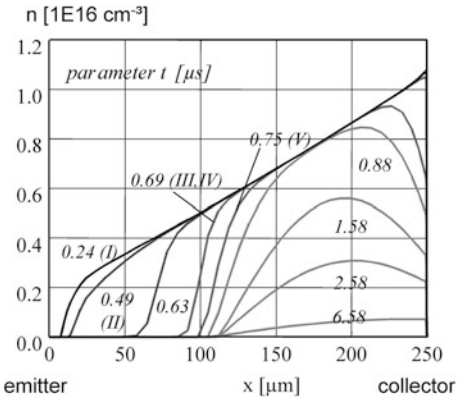


Fig. 10.10 Electron density in a 1200 V NPT-IGBT at turn-off of 80 A/cm^2. At the instant (III, IV), the device has taken the applied battery voltage, the current decreases. Figure from [Net99] © 2008 isle Steuerungstechnik und Leistungselektronik GmbH

down to the value of the tail current, similar as in Fig. 10.6. The extracted part of the base extends to approximately $x = 90$ µm. After $t = 0.75$ µs, the remaining charge on the right-hand side of the base is removed. In Fig. 10.10, this process is no more driven by the electric field. The applied voltage amounts to 600 V, the space charge has extended up to 100 µm and has taken up the voltage. There is only a very low increase in the extension of the space charge in the further process. For the removal of the carriers in the part close to the collector, recombination is the determining mechanism. Because of the high carrier lifetime, there is still a considerable charge in the device even up to $t = 6.58$ µs; during all this time t_{tail}, the tail current flows.

The voltage across the device has the value of the battery voltage during the time interval t_{tail}. Therefore, a main part of the turn-off losses is created by the tail current.

The PT-IGBT shows a shorter, faster decaying tail current because of its lower width of the base layer. However, PT-IGBTs have a more hanging-down plasma distribution in the conduction mode, caused by the reduced carrier lifetime. Compared to Fig. 10.9, the plasma density is lower at the emitter side, but higher at the collector side. As a result, the tail current in PT-IGBTs is shorter, but higher.

At the turn-off process described before, in which first the voltage increases before the current decays - switching with inductive load and also assigned as “hard switching” - the turn-off losses in the PT and NPT-IGBT are similar. However, the long tail current of the NPT-IGBT is a disadvantage in a turn-off process of the type “soft switching”, at which at a circuit-induced zero crossing of the voltage or close-to-zero crossing the current is turned off, and the voltage increases slowly. Only a low share of the carriers is then removed in the first phase of the turn-off process. During the tail current time, the voltage increases and can lead to an additional increase in the current at the time t_{tail}. With this, additional losses are created.

10.6 Modern IGBTs with Increased Charge Carrier Density

The first IGBT-generations had a plasma distribution as shown in Fig. 10.9. This distribution is expected for a bipolar pnp-transistor, compare Fig. 7.6. Therefore, it was first supposed that the IGBT will have similar limits as a bipolar transistor: high voltage drop V_C in the on-state, and purely suitable for a blocking voltage above 1700 V. However, the IGBT could be improved.

Figure 10.10 shows that the plasma stored at the emitter side of the IGBT is quickly removed by the electric field while the voltage grows at the turn-off process. The plasma distribution, which is high at the collector side, low at the emitter side (see Fig. 10.9), leads to the effect that the main part of the charge carriers is removed in the tail phase. At the emitter side, the density of charge carriers could be significantly enlarged, without much increase in the turn-off losses. A higher density of carrier plasma will lead to a reduction of the voltage drop V_{drift} in the base layer, and consequently to a lower on-state voltage drop V_C. To achieve this, it was supposed during a long time that a new device will be necessary. Research and development activities on MOS-controlled thyristors (MCTs) and similar devices were started. However, it was discovered that the IGBT is capable to achieve the desired internal profile of the carrier plasma, and that the new devices might not be necessary.

10.6.1 Plasma Enhancement by High n-Emitter Efficiency

The effect of a possible increased plasma density at the emitter side was shown by Kitagawa et al. in 1993. They designated such a device as "Injection Enhanced Insulated Gate Bipolar Transistor (IEGT) [Kit93]. Kitagawa et al. discovered the effect at a Trench-IGBT, designed for 4.5 kV. The device had an increased charge carrier density at the emitter side and a surprisingly low voltage drop at forward conduction. The principal effect can also be explained with a planar IGBT, however [Lin06].

Figure 10.11 shows a partial section of an IGBT structure. In a first simplified investigation, the IGBT can be divided in two areas: an area of a bipolar pnp-transistor, and an area of a pin diode.

In the *transistor area*, the IGBT behaves as a pnp-transistor in the saturation region. The collector side of the IGBT corresponds to the emitter side of the pnp-transistor, see Fig. 10.2. The density of free carriers is high at the junction J_1 and decreases towards the junction J_2; at the junction J_2 it approximates to zero. For comparison, see Chap. 7 on the bipolar transistor (Fig. 7.6) and its description. From this, a shape of the carrier plasma as drawn in Fig. 10.9 results. In this figure, the junction J_2 is located at $x \approx 7$ µm. This shape of the plasma holds if it is determined by the bipolar transistor.

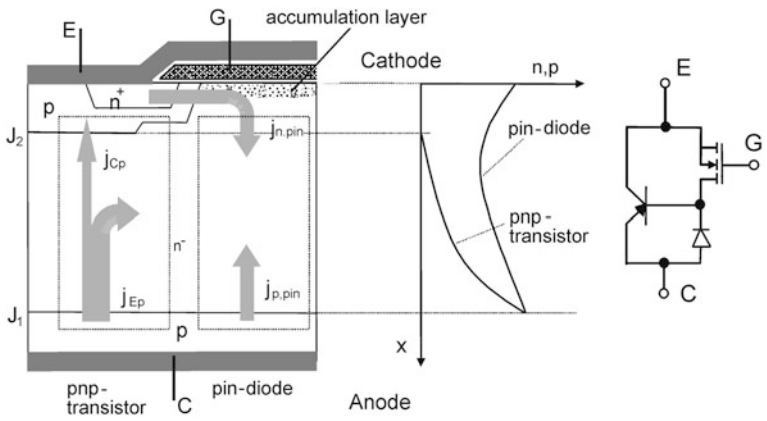


Fig. 10.11 Partition of an IGBT in a pnp-transistor area and a pin-diode area, carrier plasma distribution in both areas, equivalent circuit containing the pin-diode with MOS-switched n-emitter

In the diode area beneath the gate, similar conditions are given in the IGBT as in a pin diode. The electrons are supplied by the MOS channel. The MOS channel behaves as an ideal emitter; the current at this point is pure electron current. At the semiconductor surface between the p-wells, an accumulation layer of electrons will build up, created by the positive voltage at the gate above this area. Holes coming from the junction J_1 will not find a path here. The plasma distribution in this area will approximate the plasma distribution of a pin diode, as it was described in Chap. 5 with Fig. 5.6 and its discussion.

The voltage drop in the diode area, which is - in first approximation - the voltage drop V_C of the IGBT in forward conduction, is given by

$$V_C = V_{Ch} + V_{drift} + V_{J1} \tag{10.15}$$

whereby V_{Ch} is the voltage drop across the channel, V_{drift} is the voltage drop in the base layer and V_{J1} is the diffusion voltage of the pn-junction J_1. To make V_{drift} low, the relation of diode area to pnp-transistor area should be as high as possible. This can be achieved if the cells have a high distance. V_{drift} will decrease with increased cell distance (cell pitch). But then the cell density of the IGBT is decreased, and therefore also the voltage drop across the channel V_{Ch} will increase, as explained in the Chap. 9 on the MOSFET. For the planar IGBT in Fig. 10.11, one finds a cell distance at which the result for V_C has a minimum.

A more detailed investigation is given by [Omu97]. The whole top-side cell structure is considered to be an n-emitter. In the area between the p-wells, beneath the gate oxide, an accumulation layer of free electrons is built up, caused by the positive voltage at the gate. This accumulation layer is summarized with the p-wells to an effective n-emitter. An n-emitter of high efficiency shall be created. For this, the emitter must inject a high electron current. For the efficiency of an n-emitter, results are in analogy to (3.99)

$$\gamma = \frac{j_n}{j} \tag{10.16}$$

Here, j_n is the electron current delivered by the channel. Equation (10.16) can be written in a new form if $j = j_n + j_p$ is used

$$\gamma = \frac{j - j_p}{j} = 1 - \frac{j_p}{j} \tag{10.17}$$

To achieve a high γ, it is necessary

(a) to increase the distance between the cells. In [Omu97] it is shown that the share of j_n at the total current increases with increasing cell distance.
(b) to take care that j_p is only a small share of the total current density j. The hole current j_p flows across the p-well, see Fig. 10.11. If the area of the p-wells is reduced, also j_p is reduced. In the planar structure in Fig. 10.11, this happens by reduction of the p-well area.

So it is possible to increase the density of the plasma and therewith to decrease the voltage drop V_{drift} if the flow of the hole current j_p is reduced. The plasma distribution close to an emitter can be strongly influenced by its efficiency. On the example of the MOS-Controlled Diode (MCD), we have seen in Fig. 5.39 that the voltage drop in the drift layer can be controlled by the emitter injection. There, in the same way, we had an effective emitter efficiency of the p-well *and* the channel. In the MCD, the combination forms a p-emitter, and the current via the n-channel is the minority current. By injection of minority carriers from the channel, the emitter efficiency was reduced with the aim to reduce the plasma concentration at the side of the p-emitter. In the IGBT, the p-well, the channel and the electron accumulation layer form an n-emitter. The holes flowing from the p-well are the minority carriers. Reducing the minority carrier current j_p will increase the efficiency of the n-emitter; with this, the plasma density close to the emitter is increased. More charge carriers are available for current transport in the wide base layer, and the voltage drop V_C is decreasing.

This principle can be realized with planar structures and with trench structures. The trench structure offers special advantages for this. An example is the Infineon trench IGBT [Las00] . The trench cell is shown in Fig. 10.12. Emitter layers of the n^+-type and channels are arranged only in the middle of the cell between the two trenches. Outside the cell, a layer without contact to the emitter is arranged. Compare the IGBT trench cell with the MOSFET trench cell in Fig. 9.6. The mode of action of the trenches is different for the IGBT and for the MOSFET. At the MOSFET, a high part of the surface had to be equipped with channels, since the channel resistance R_{Ch} determines a main part of the total voltage drop; it is made low by arranging many channels in parallel. In the IGBT, the channel resistance is of minor effect to the voltage drop in the on-state. This voltage drop in the IGBT is determined by the plasma density in the middle layer, since the IGBT is a bipolar device. This plasma density must be high to reduce the voltage drop. Therefore, one has to reduce j_p to increase γ.

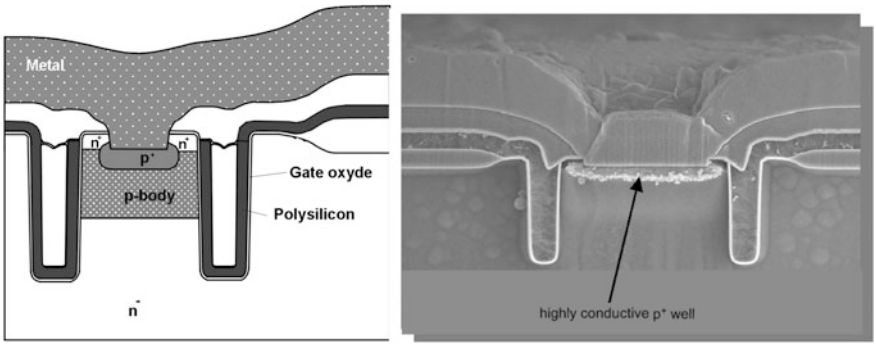


Fig. 10.12 IGBT trench cell. Structure (left), picture of a cross section of a cell made with a raster electron microscope (right). Figures from T. Laska, Infineon Technologies

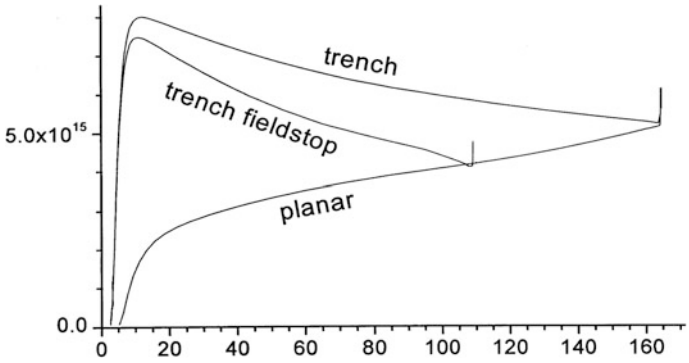


Fig. 10.13 Plasma distribution in a conventional IGBT (planar), in a trench-IGBT and in a trench fieldstop IGBT. Infineon figure [Las00b]

A comparison of the plasma density in the trench IGBT with the conventional NPT-IGBT according to Fig. 10.8 is given in Fig. 10.13 [Las00b]. The emitter is on the left-hand side of Fig. 10.13, the collector is on the right-hand side. The line for "planar" is the conventional IGBT, there the plasma distribution is as expected for a pnp-transistor with its emitter on the right-hand side. With the trench IGBT, we get an enhanced plasma concentration at the IGBT emitter side. It approximates the plasma distribution in a pin-diode.

In [Tak98], it was even shown that the trench IGBT voltage drop V_C decreases if some of the cells are not contacted. The plasma density increases in the same amount, as the part of contacted cells is reduced, as long as the number of not contacted cells is not too high. V_C is reduced, because the effect of plasma enhancement is much higher than an increased voltage drop across the channel.

10.6.2 The "Latch-up Free Cell Geometry"

Essential for modern IGBTs is to avoid latch-up of the parasitic thyristor at turn-off. This requirement must now be met under the condition of increased plasma density. The cell must have a structure which is optimized for this requirement. Figure 10.14 shows a detail of a trench cell. The hole current will flow mainly close to the electron current because of the condition of neutrality. Therefore, it flows close to the channel, and in its further way to the contact at the p^+-region it must flow underneath the n^+-source region. The n^+p-junction is biased in forward direction. If the voltage drop V_p that is generated by the hole current below the n^+-layer across the length L reaches the order of magnitude of the built-in voltage V_{bi} of this n^+p-junction, then the n^+-region will inject holes. The consequence will be latch-up of the parasitic thyristor, the turn-off capability is lost, the control of the device is lost and destruction of the IGBT will follow.

Using the depictive representation in Fig. 10.14, this voltage drop V_p can be simplified according to [Ogu04] with

$$V_p = \int_0^L R \cdot j_p \cdot y \cdot dy = \frac{1}{2} \cdot R \cdot j_p \cdot L^2 \tag{10.18}$$

where R is the sheet resistivity of the p-layer below the n^+-source in $\Omega/\square$ and L is the length of the source layer. To keep V_p below the built-in voltage (≈ 0.7 V at 25 °C and decreasing with temperature) even at very high current density, especially L must be small, but also ρ must be kept low. As one can recognize in Fig. 10.12, L is designed very short in modern trench IGBTs. A highly doped p^+-layer spreads as far as possible in direction of the trench below the source layer. With these design measures, destructive latch-up can be avoided even at high current density.

This design measure, termed to be a "latch-up free cell geometry" [Las03] is realized by a highly conductive p^+ well adjusted on a submicron scale (Fig. 10.14) concerning the distance of this p^+ well to the trench sidewall. This layer forms a

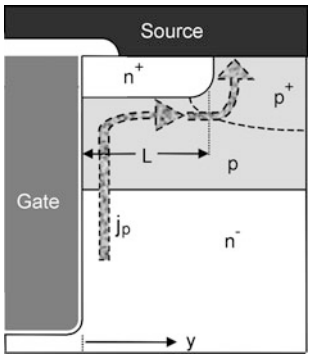


Fig. 10.14 Detail of an IGBT trench cell

resistance R_S (see equivalent circuit Fig. 10.2b). If R_S is low enough, the parasitic npn bipolar transistor is effectively suppressed, and latch-up of the thyristor, containing both the npn- and pnp bipolar transistors, will not occur. The discussed measure is not restricted to the trench cell. The same measure is possible for a planar structure.

We will come back on this later in Chap. 13. In devices for high blocking capability, a mode of operation with dynamic avalanche occurs at turn-off, an additional hole current is generated and the current density may even be locally increased. IGBTs with well-designed cell structures overcome even these high stress conditions.

10.6.3 The Effect of the "Hole Barrier"

The possibility to increase the plasma density below the emitter cells is also given by the implementation of an additional n-doped layer. This was first shown by [Tai96] on the example of a trench IGBT, the structure was denominated as "Carrier Stored Trench Gate Bipolar Transistor" (CSTBT). The effect is explained in [Tai96] in the following way: at the n^-n^+-junction a diffusion potential of approximately 0.17 V is built up, it hinders the outflow of holes. This additional layer was designated as a hole barrier.

This n-doped layer below the p-well acts in the same way in a planar IGBT [Mor07, Rah06], as shown in Fig. 10.15. In the n-doped layer, the hole current is the minority carrier current. It decreases strongly before it enters the p-well. In Eq. (10.17), this reduces j_p and increases the emitter efficiency γ of the total top-side cell structure, in which we again summarize p-well, n-channel and n accumulation layer to an effective n-emitter. The consequence is an increased plasma density, as shown in Fig. 10.15, right-hand side.

The hole barrier hinders the outflow of holes. To hold the condition of neutrality, additional electrons are delivered by the n-channel. The density of plasma increases, see Fig. 10.15, right-hand side.

A disadvantage of this measure is that the increased doping below the blocking junction J_2 decreases the blocking capability. This must be compensated with a slightly increased thickness of the n-base of the IGBT, and this increases the forward voltage drop. A part of the achieved advantage is lost again, but this effect can be kept minimal [Lin06].

A combination of the hole barrier with the trench structure is done in the "Carrier Stored Trench Gate Bipolar Transistor" (CSTBT) of the manufacturer Mitsubishi (see Fig. 10.16). The n-layer as a hole barrier is arranged below the p-layer within the trench structure. Below the hole barrier, holes accumulate; to keep neutrality, additional electrons are delivered effectively by the channel. The density of plasma is increased locally.

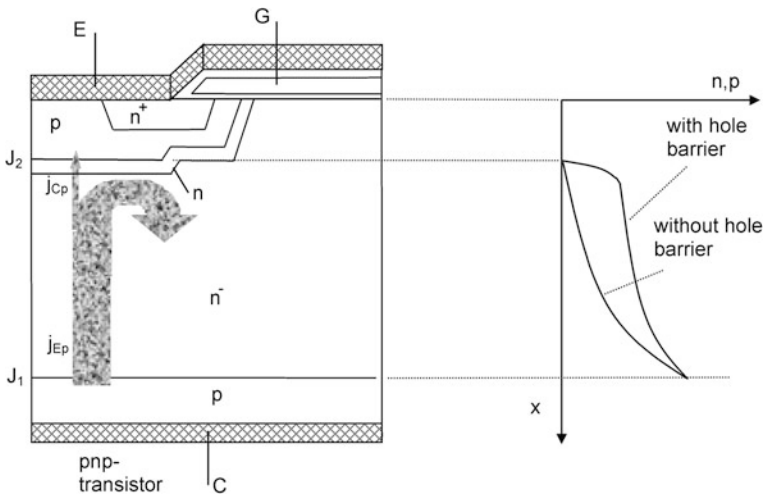


Fig. 10.15 Increase of the density of free charge carriers by a hole barrier

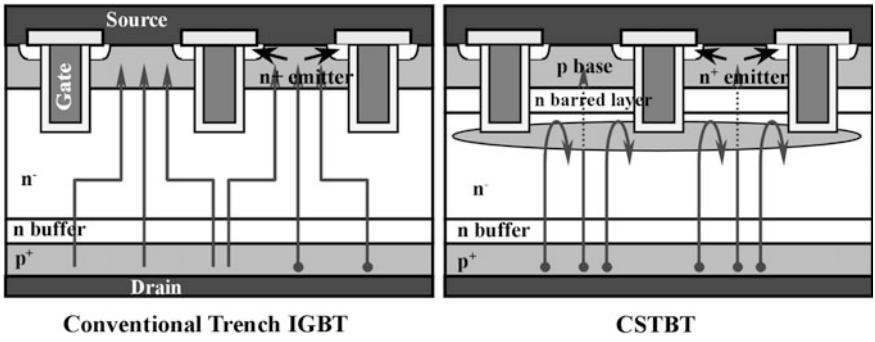


Fig. 10.16 Carrier Stored Trench Gate Bipolar Transistor (right-hand side) in comparison to a conventional trench IGBT (left-hand side). Pictures from Mitsubishi Electronics

The hole barrier can be combined with the before-described measures of increasing the distance between the cells and decreasing the lateral extension of the p-layers. In a trench IGBT, this can be done very easily by debarring a part of the cells. These cells are denominated as "plugged cells" [Yam02]. The polysilicon in these cells, which forms the gate area, is shorted to the emitter metallization, and the cell is prevented from building an n-channel. This measure additionally has the advantage that the current I_{Dsat} in the active area is reduced, and therewith the current at a short circuit event is reduced.

10.6.4 Collector Side Buffer Layers

Besides the described increasing of the plasma density, every modern IGBT additionally uses the effect of a reduced width of the n-base layer by designing a trapezoidal electric field instead of a triangular one. This is achieved with an n-layer with increased doping in front of the p-collector layer. The denomination differs for different manufacturers: "fieldstop", "soft punch through", "light punch through", etc., but it is more or less the same.

The plasma density for the "trench fieldstop" IGBT has already been shown in Fig. 10.13. In a trench-fieldstop IGBT, the width of the n-base is shortened compared to a trench IGBT. Figure 10.17 shows the new structure. In front of the collector layer, the doping density is increased (n-fieldstop). The space charge is trapezoidal if a voltage close to the specified blocking voltage is applied. This is sketched in Fig. 10.17, left-hand side. In reality, it is only a moderate trapeze, it is closer to a triangle and not almost rectangle, as one could conclude from Fig. 10.17.

From the viewpoint of the shape of the electric field, the fieldstop IGBT is similar to a PT-IGBT. The base width w_B is significantly reduced at the same blocking voltage in an IGBT with collector side buffer layer compared to an NPT-IGBT. The voltage V_{drift} that drops across the base layer is proportional to w_B^2 according to Eq. (10.13). Hence the voltage drop V_C can be reduced significantly with this design.

However, this is the only common feature with the PT-IGBT. The trench-fieldstop IGBT is not fabricated like the PT-IGBT on a p^+-substrate with an epitaxial n-layer, it is rather fabricated from a homogeneously n-doped wafer. The fabrication of a collector layer with exactly adjusted emitter efficiency is done similarly as with an NPT-IGBT [Las00b]. α_{pnp} is adjusted by the emitter efficiency; an emitter of low penetration depth and low doping is used. No charge carrier lifetime reduction is done. The mode of function of the trench-fieldstop IGBT is closer to the NPT-IGBT than to the PT-IGBT. As a result, the temperature dependency of V_C at is similar to that of the NPT-IGBT. The desired "positive temperature coefficient" remains.

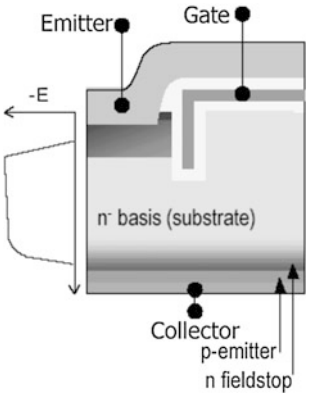


Fig. 10.17 Structure of the Infineon trench-fieldstop IGBT. Figure from Infineon

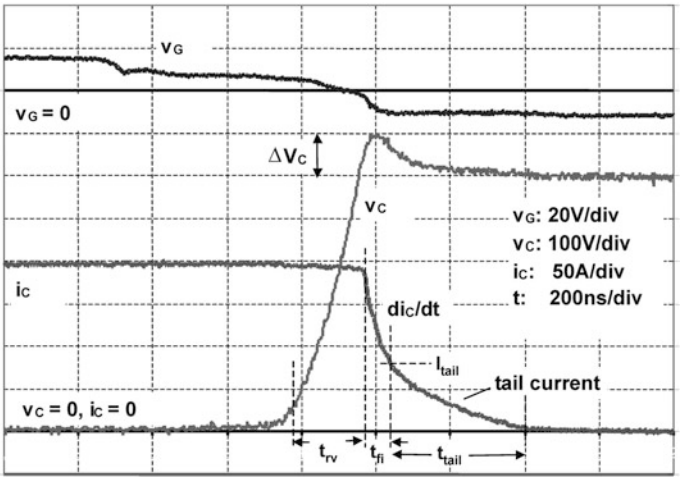


Fig. 10.18 Turn-off of the trench fieldstop IGBT (FF200R12KE3 from Infineon, 200 A 1200 V module). T = 125 °C, $R_{Goff} = 5\,\Omega$

Figure 10.18 shows the turn-off behavior of the trench-fieldstop IGBT. If one compares this figure with the turn-off behavior of the NPT-IGBT in Fig. 10.6, one can see a significantly shortened tail current. Additionally, the fall time of the current t_{fi} is increased and di_C/dt is decreased; this leads to a lower voltage peak ΔV_C.

The shortening of the tail current happens by the effect that a lower part of the stored plasma in the on-state remains close to the collector side at turn-off. The removal of the stored charge is done in a big amount during the voltage rise time t_{rv}. If the applied voltage V_{bat} is increased above the voltage of 600 V, as applied in Fig. 10.18, the tail current is shortened further, and finally it can disappear completely. Then the space charge spreads across the whole middle layer. If V_{bat} is increased further, a current snap-off similar to the behavior at reverse recovery of a snappy diode will occur at the end of the fall time t_{fi}. This current snap-off can lead to high overvoltage peaks.

10.7 IGBTs with Bidirectional Blocking Capability

For some applications of power electronics, e.g. the matrix converter, a both-side blocking device is necessary. The IGBT structure contains a p-layer forming a pn^--junction at the collector side. The basic structure, as drawn in Fig. 10.1, has the capability to take an electric field at the bottom-side pn^--junction as well as at the top-side n^-p-junction. It has similarities to the structure of a thyristor which has a blocking capability in both directions. However, the back-side pn^--junction has no

defined junction termination. Such planar junction terminations for shallow pn-junctions need microstructures. The processes for the fabrication of junction terminations are only possible at the top side of the wafers, since semiconductor technology has been optimized for microstructures on one side of a wafer.

A possible solution is to lead the back-side pn-junction to the front side of the wafer. This can be done by the diffusion of a deep p-layer which reaches through the whole wafer; this technology is called "diffusion isolation" [Tai04]. A schematic drawing of an IGBT with this deep edge diffusion is shown in Fig. 10.19. With a deep diffusion in the area where the wafer is cut in the last production step, the collector-side pn^--junction is connected to the front side of the wafer. Now it is possible to apply a junction termination for both directions. This is done in Fig. 10.19 for the forward direction with potential rings, as known from Fig. 4.20. The junction termination for the forward direction ends at the channel stopper. As junction termination for the reverse direction, a field plate structure is applied (see Fig. 10.19).

The device blocks the voltage in both directions like a thyristor. The bidirectional blocking IGBT must be dimensioned as NPT-type. A buffer layer in front of the p-collector would reduce or even eliminate the blocking capability in reverse direction. Therefore, a minimal thickness w_B of the base layer for a triangular field shape is required. In [Nai04], this is given with 200 µm for a 1200 V reverse blocking IGBT; the electric field in the volume is very similar to that in Fig. 10.8.

Since a buffer layer cannot be applied, the reverse blocking IGBT cannot have a forward voltage drop V_C as low as other modern IGBTs. But V_C is surely lower than the one of a serially connected IGBT and diode.

In the intended matrix converter applications, the reverse blocking IGBT works as freewheeling diode at some switching events; it is passively turned-off like a diode. Because of the high carrier lifetime in IGBTs, a high reverse recovery peak I_{RRM} and a high reverse recovery charge occur. In one concept [Nai04], electron irradiation was used and a reduction of I_{RRM} of about 10% was achieved. The shorter carrier lifetime increases the on-state losses; a trade-off must be made.

Further work is done to improve the reverse blocking IGBT. The deep diffusion on the right-hand side of Fig. 10.19 has, however, also the consequence of lateral diffusion which will be approximately spread by 0.8× of the diffusion depth to the

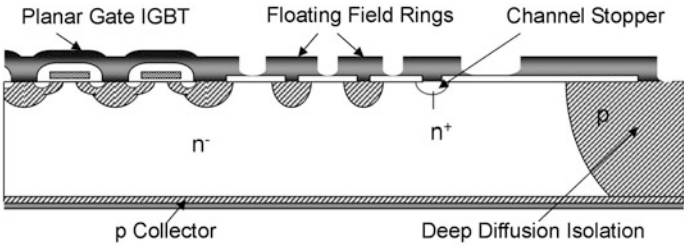


Fig. 10.19 Symmetrically blocking IGBT with edge diffusion. Figure from [Ara05] © 2005 EPE

side. With a wafer thickness in the range >100 µm, this deep p-layer will also be very wide. This leads to a loss of area for the region which takes the current, a loss of active area, and a high share of area for the junction termination which cannot be used for current transport. Research and development work is done for improved structures, for example to replace the deep diffusion zones by deep trenches [Ara05].

10.8 Reverse Conducting IGBTs

If in a part of the collector area an n^+-layer is implemented, see Fig. 10.20, a diode is formed by this n^+-layer and the p-body. There are several advantages of the monolithic integration of the diode. Primarily, it is the reduction of the chip size. Thermal calculations in of the diode and the IGBT in a three-phase inverter application [Rut07] have led to the result that a great saving of the area is possible, while the RC-IGBT stays almost at the same size like the IGBT and the diode area can be saved.

The concept of an RC-IGBT in a productive volume was first realized with an optimization for lamp ballast applications [Gri03]. This application needs 600 V blocking voltage, a current in the range of 1 A, and there is no hard commutation of the diode. These devices could be realized successfully without optimization of the diode.

To satisfy the requirements of hard switching applications, the reverse recovery behavior of the integrated diode has to be optimized. For this diode, the n-doped regions on the backside act as a cathode emitter (see Fig. 10.20), while the p-body of the IGBT and the highly p-doped anti-latch-up region near the front side act as an anode emitter of a freewheeling diode integrated into the chip in this way.

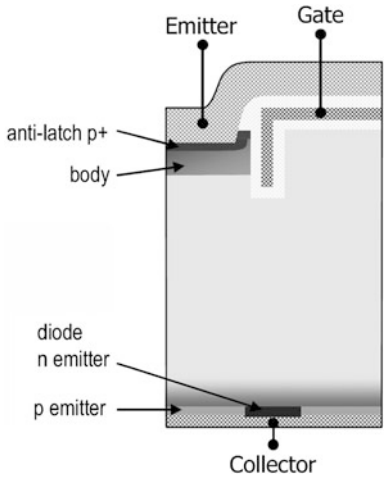


Fig. 10.20 Reverse conducting IGBT with trench gate. Figure from [Rut07] © 2007 IEEE

Unfortunately, such a diode with a highly doped p-emitter has a plasma distribution as shown in Fig. 10.21. It is very high at the anode side and lower at the cathode side. This will lead to a high reverse recovery peak and snappy reverse recovery behavior, as discussed in detail in Chap. 5. The internal plasma distribution as it must be for a soft recovery diode is shown for comparison in Fig. 10.21, too.

To reduce the plasma density at the cathode side, different methods have been tested. Irradiation of He^{2+}-ions is applied for the reverse conducting IGBT in [Tai04b]. The reverse recovery current peak I_{RRM} was reduced close to the value of a commercial freewheeling diode. But irradiation technologies can have a disadvantage. In [Rut07], it is reported that particle irradiation causes also undesirable trapped charges in the gate oxide and the silicon-gate oxide interface. These charges give rise to a drop of the threshold voltage and a broader parameter distribution. Therefore, the way used in [Rut07] is to reduce the efficiency of the anti-latch p^+-emitter by reducing the implantation dose. The advantage of this method is that there is no effect on the V_C. However, a reduction of this doping dose is limited by the overcurrent turn-off robustness (latch-up robustness). It was found that the potential for a reduction of Q_{RR} by decreasing the p^+-emitter dose at the same robustness as today's devices is between 15 and 25% [Rut07]. With platinum diffusion, Q_{RR} was reduced by 60%, and the increase in V_C of the IGBT was only 0.1 V [Rut07].

The measures to improve the IGBT and to improve the freewheeling diode contradict each other in some aspects. However it is expected that reverse conducting IGBTs will become available also for switching with inductive load, since a reduction of costs in the power electronic system is possible by saving half of the number of power dies. These solutions might be successful in applications in which the requirements to the diode reverse recovery are medium, e.g. for low current

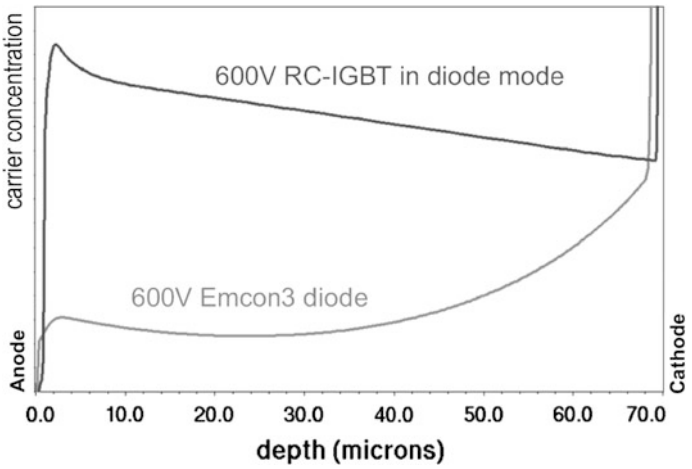


Fig. 10.21 Carrier concentration profile in a 600 V EMCON3 compared to a 600 V RC-IGBT in diode mode. Figure from [Rut07] © 2007 IEEE

applications up to 1 kW, since the energy stored in the parasitic inductance is small at low current. There is a wide field of such applications, such as inverters in air conditioners, refrigerators, and others.

A further interesting concept for a reverse conducting IGBT rated 3300 V was presented in [Rah08]. The requirements to the freewheeling diode are highest in the intended high power motor drive application. The new structure is shown in Fig. 10.22.

This structure uses the fact that there are measures which are not conflicting in their effect on the plasma distribution as well in the IGBT as in the diode. An n-layer below the p-well is used, as it was named as "hole barrier" in Fig. 10.15. It decreases the injected hole current, for the effective emitter of the IGBT it acts in an enhanced emitter efficiency. For the diode, it acts as reduced p-emitter efficiency.

On the collector side, the n^+-layer acts as a kind of anode short, known from GTO-thyristors, it will reduce the IGBT collector side emitter efficiency. Doping of the collector side p-layer, as well as width and doping of the n^+-layer must be carefully adjusted to fulfill the requirements for the diode as well as for the IGBT. For the diode, additional a local lifetime control region below the p-well is applied, as known from the CAL-diode in Chap. 5. The generation of recombination centers is executed with masked irradiation of light ions. The reverse recovery of the diode can further be improved by opening the channel. Injected electrons by the channel in the diode conducting mode will reduce the p-emitter efficiency, as it is described on hand of the MOS controlled diode (see Sect. 5.7).

Investigations in [Rah08] show that the IGBT as well as the diode could even profit in some aspects due to the integration. For the application it is a high advantage. The effective IGBT area is increased by 50% and the effective diode area even by 200%. A potential for rating 50% higher current for IGBT in modules with the same area is given, however only a part of it can be used in practice because of the thermal resistance of the housing sets a limit.

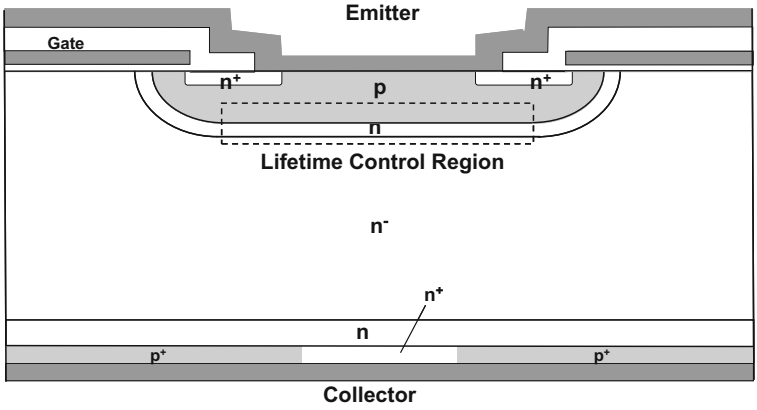


Fig. 10.22 3.3 kV RC-IGBT with optimized integrated diode. Figure following [Rah08] © 2007 IEEE

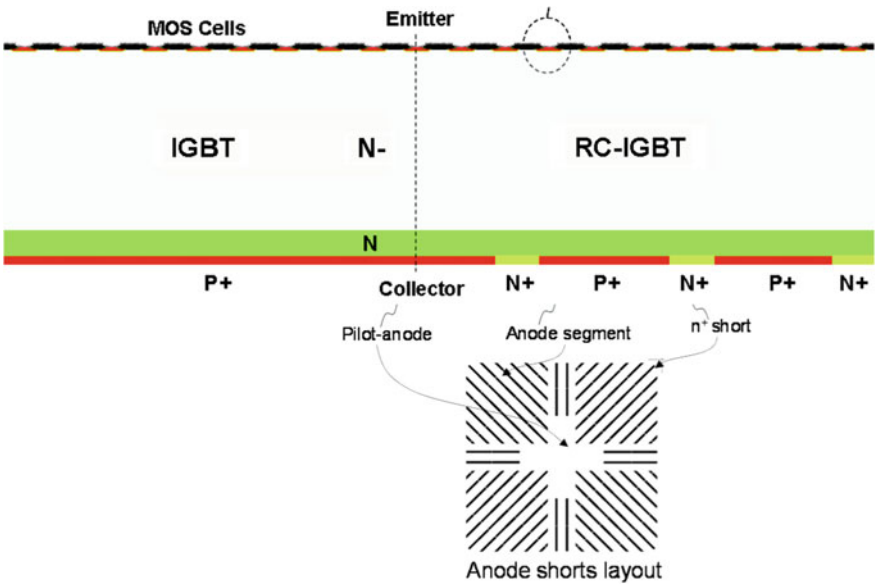


Fig. 10.23 Bimode-Insulated-Gate-Transistor BIGT. Fig. adapted from [Rah09] and [Sto15]

Realized RC IGBTs, however, showed a snap-back phenomenon in in the forward characteristics: First the n-region carry the current. As the p-layers starts to inject carriers, the forward voltage snaps back to a lower voltage. To avoid this unwanted behavior, the device is divided into an area with usual IGBT with no n-shorts (pilot IGBT) and an area with reverse conducting possibility. The device was denoted as Bimode-Insulated-Gate-Transistor BIGT [Rah09]. It is shown in Fig. 10.23. The optimal design of the n-shorts was found to be in the form of radial stripes [Sto11], an approximation to this is shown in Fig. 10.23 on the bottom side.

A 6.5 kV reverse conducting IGBT, where the freewheeling diode is integrated into the IGBT, was presented in [Wer14, Wer15]. The device is denoted as "Reverse Conducting IGBT with Diode Control" (RCDC) (Fig. 10.24).

In transistor operation the former area of the diodes is additionally used, during diode operation additionally the former transistor area. With this integration, a module with the same outline has now a rated current of 1000 A compared to 750 A of the former version. Especially interesting is the capability to strongly control the diode characteristics by the gate. In diode conduction mode, the gate is optimally loaded with −15 V. The flow of electrons is stopped, the p-emitter efficiency becomes high. The internal plasma is shown in Fig. 10.25. The voltage drop in conduction mode becomes very low, in the range of only a little bit above 2.5 V, see Fig. 10.26. Short before turn-off of the diode, a "desaturation pulse" of +15 V is applied during, for example, 15 µs. An n-channel injecting electrons is formed parallel to the p-doped diode emitter region forms. The p-emitter efficiency now becomes very low, the forward voltage is high (Fig. 10.26). The internal plasma shape of carriers becomes optimal for turn-off with low reverse recovery current

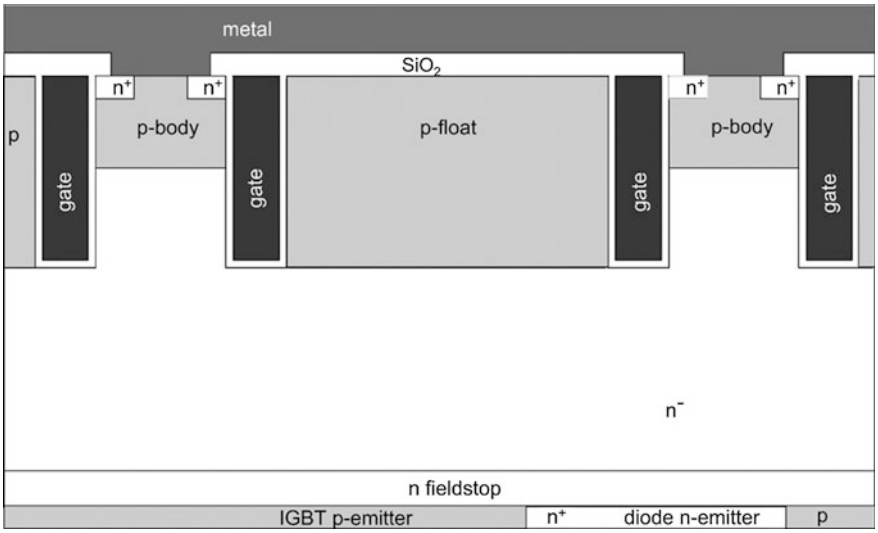


Fig. 10.24 Reverse conducting trench IGBT with integrated diode. Adapted from D. Werber, Infineon [Wer15]

maximum (Fig. 10.25). After a short dead time, the IGBT is turned-on and the diode is commutated in reverse direction. This leads to low turn-off losses of the diode (reduced by 40%) and low turn-on losses of the IGBT (reduced by 34%). The function of a field-controlled diode is shown in an impressive way.

A further advantageous aspect of the reverse conducting IGBT is expected in respect to reliability. In the typical motor drive application conditions, in one sinus half-wave the IGBT, in the other the diode is heated up. The absence of inactive periods can reduce the temperature ripple and will increase the lifetime of the module. For details, see Chap. 12. In 2017, Reverse conduction IGBTs are available for 4500 V [Dug17] and 1200 V [Tah16, Osa17]. The reverse conducting IGBT is still object of research and development.

10.9 The Potential of the IGBT

It has been shown that with modern IGBTs an improved distribution of the internal plasma was achieved. Therefore the IGBT applications could be extended to voltage ranges which were formerly only possible with thyristors. The development of devices with new structures, especially of a MOS-controlled thyristors and derivatives as IGBT-followers, has been placed back or suspended since the new structures were not necessary. The desired advantage has been possible with the IGBT to a great extent. IGBTs are commercially available up to 6.5 kV blocking

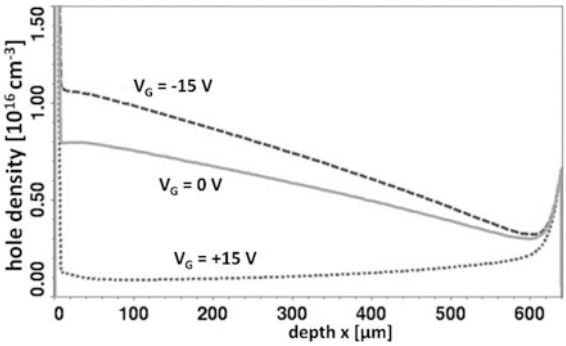


Fig. 10.25 Simulated carrier concentration of the RCDC in diode conduction mode along a vertical cut through a backside region with n-emitter for different gate voltages at 125 °C. Fig. from [Wer15], PCIM Europe 2015

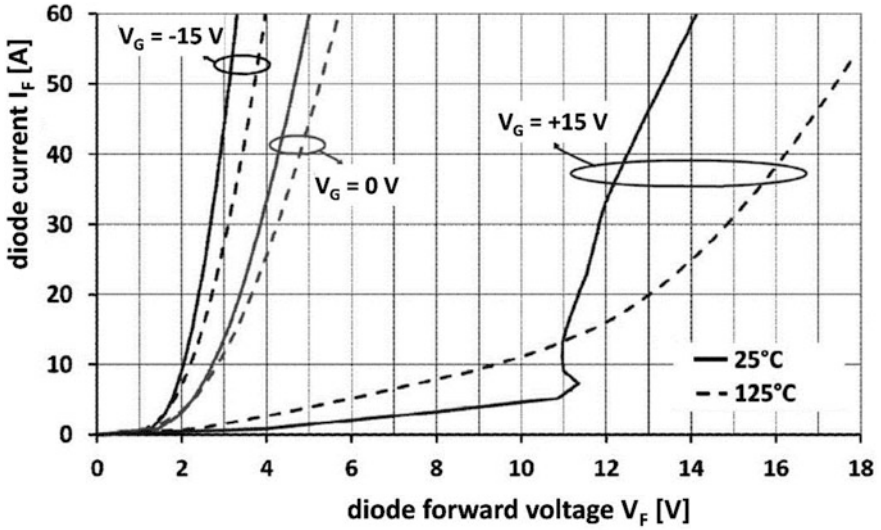


Fig. 10.26 Measured RCDC characteristics of diode conduction mode for gate voltages $V_G = -15$, 0, +15 V at room temperature and 125 °C. The nominal current per chip amounts to 28 A. Fig. adapted from [Wer14] and [Wer15]

voltage since more than ten years. Development works on IGBTs for the voltage range 8 or 10 kV were done, however these devices did not come to market.

However, further intensive work in research and development has to be done to decrease the conduction and switching losses of IGBTs. With IGBTs very low forward voltage drop and thereby very low conduction losses can be achieved [Sum12, Sum13]. A new structure as example is shown in Fig. 10.27. In the structure in Fig. 10.27a the current flowing from collector C to emitter E crosses the narrow Mesa between gate G and control gate CG. If a positive voltage is applied at CG, the hole current flow is suppressed. The structure approximates to an ideal n-emitter with j_p approximating 0 and $\gamma \rightarrow 1$. With this high emitter efficiency, the carrier density at the n-emitter side can be made very high, see Fig. 10.27b. The

on-state losses then will be low. With the signal at the control gate, the turn-off losses can be controlled. With a negative voltage at CG, the carrier density will be low and the turn-off losses are reduced. The achieved progress compared to the "IGBT-limit" calculated by Nakagawa [Nak06] is to be seen as a dot in Fig. 10.28.

The IGBT-limit was calculated in [Nak06] for an IGBT with ideal n-emitter of efficiency $\gamma_n = 1$. The lines for IGBTs in Fig. 10.28 are calculated as $R_{on} = V_C/I_C$ at the operation point of rated current, this means they are simplified for bipolar devices. In the publication [Sum13], for a 1.2 kV IGBT at 300 A/cm^2 a forward voltage drop of 1.6 V is achieved, this is even close to the forward voltage of a thyristor rated for this blocking voltage.

In HVDC applications, today IGBTs with rated voltage of 3.3 and 4.5 kV are used. 6.5 kV and more could be the future voltage rating to need less devices in series connection for the output voltage of the converter which is 500 kV or even more. A 6.5 kV IGBT of today has a V_C in the range 3.7 V. This is much higher than the value theoretically possible at the Nakagawa-limit.

An exemplifying drawing (T = 125 °C) for a state of the art 6.5 kV is shown in Fig. 10.29. A state of the art module (2016) is rated to 750 A and has a voltage drop of 3.7 V at rated current and 125 °C. It contains 24 IGBT-Chips with an approximated active area of 1.21 cm^2 each. This results in a R_{on}·A of 143 mΩcm^2 (see Fig. 10.28, "actual Si-IGBT"). At the "IGBT-Limit" according [Nak06], 7.5 mΩcm^2 would be possible. It is assumed that one can approximate to this limit up to a factor of two meaning 15 mΩcm^2 are achieved. This leads for same power loss density to a current conduction capability of 2300 A and a forward voltage drop of 1.45 V.

If this aim would be reached, the number of IGBT modules for a power converter with same output power could be reduced down to one third, and, compared to a solution with state of the art IGBTs, only 15% of conduction losses would be dissipated. However, an optimization must consider all requirements, especially overload capability and short-circuit withstand capability. Additionally, the switching losses have been neglected in this consideration. However, some high-power applications, for example the Modular Multilevel Converter for HVDC applications, use low switching frequencies. Therefore, such an aim is not unrealistic.

Additionally, IGBTs for the most frequently used 1200 V voltage range have a high capability of reduction of losses. It is achieved that IGBTs with a further reduction of the forward voltage down to less than 1.5 V will be possible. The losses in inverters for motor control can be reduced strongly.

Investigating Fig. 10.28 again, it is visible that for voltages above about 3 kV IGBTs can be superior to SiC unipolar devices regarding the forward voltage drop. SiC bipolar devices are possible, however due to the deep impurity levels of acceptors, incomplete ionization takes place for highly doped p-emitter regions. There is work in research for SiC IGBTs. However it will take time until they become mature devices.

IGBTs with additional integrated functions, as the reverse blocking IGBT and the reverse conducting IGBT, are expected to become commercially available. There is significant progress in reverse conducting IGBTs.

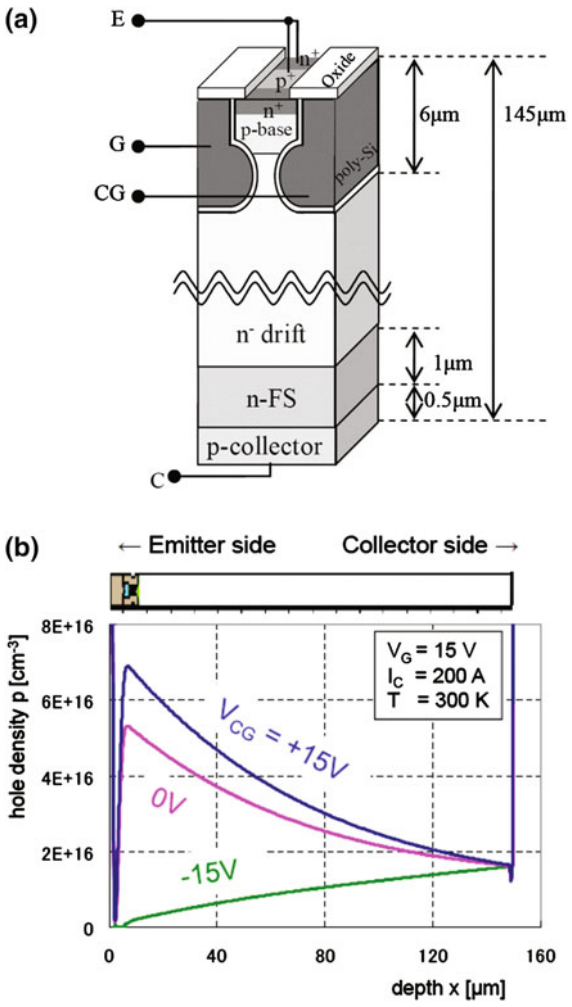


Fig. 10.27 IGBT structure with very low forward voltage drop. **a** Structure, **b** Flooding with free carriers. Figs from [Sak13] © 2013 The Institute of Electrical Engineers of Japan

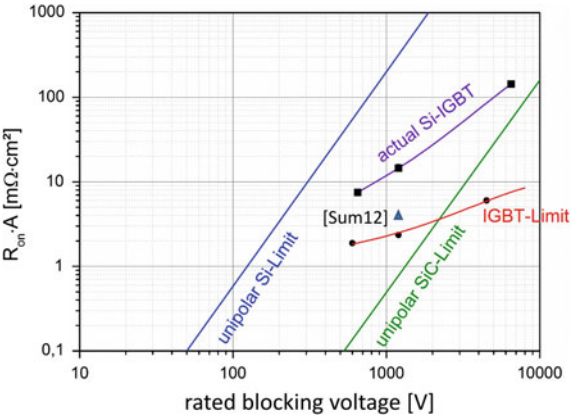


Fig. 10.28 Comparison of the on-resistance for the structure in Fig. 10.27 (marked as [Sum12]) with the on-resistance of unipolar devices (Si-Limit, SiC-Limit), with state-off-the-art IGBTs (actual Si-IGBT) and with theoretically possible IGBT-Limit, which is referred as Nakagawa-Limit. Figure inspired by [Nak06] and [Sum12]

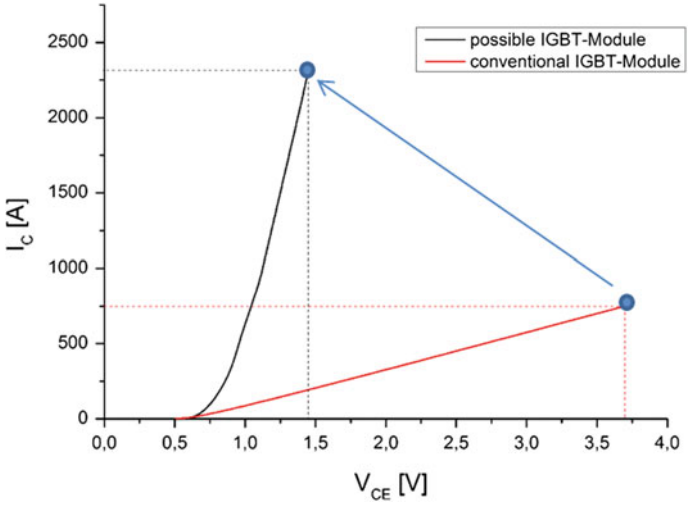


Fig. 10.29 Exemplifying drawing of the forward characteristics (T = 125 °C) for a state of the art 6.5 kV module and the possible progress at approximation to the Nakagawa-limit up to a factor of two

With the progress in IGBTs it is possible to control more and more power in a small device area. But with this, the dissipated losses per area are also increasing. These losses have to be extracted by the package and the challenges and requirements for the packaging technology are increasing. This is the subject of the next chapter.

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